

BIOMEDICAL SENSORS Advancing Medical And Biotechnology



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Sensors are small, tiny and intelligent devices that are used to measure physical variables like temperature, humidity, gas, velocity, flow rate, pressure and so on. According to American National Standards Institute, a sensor is defined as a device that provides a usable output in response to a specified measure.

A sensor acquires a physical quantity and converts it into a signal suitable for processing, for example, optical, electrical and mechanical, whereas a transducer converts a signal in one form of energy to a signal in another form.

According to the basic sensing principle, sensors can be classified into mechanical, electrochemical, bio, optical, semiconductor, magnetic and thermal types.

According to physical parameters measured by sensors, these can be classified into resistance displacement, inductive displacement, capacitive displacement, piezoelectric pressure, laser interferometer displacement, bore gagging displacement, ultrasonic displacement, optical encoder displacement, optical fibre displacement, optical beam deformation, flow, imaging, temperature, intelligent and chemical ingredient types.

The medical sensors market

The global medical sensors market is estimated to reach US\$ 15.01 billion by 2022, at a CAGR of 8.5 per cent between 2016 and 2022. People are increasingly adopting home healthcare services owing to the rising costs of medical treatments in hospitals and medical care clinics. With this, demand for various healthcare devices is expected to increase in the next few years. Introduction of new medical

sensors in the global market is expected to contribute towards the growth of the global medical sensors market in the years to come.

Market sizing for global medical sensors was done by top-down and bottom-up methods. In the bottom-up approach, market size, in terms of volume, for each type of medical sensors was calculated with their average selling price and then each type was mapped with applications. Summation of revenue from all types of sensors and applications gave the overall market size.

In the top-down approach, top companies in the market were analysed and their product portfolio studied to get the global market size for medical sensors. This overall market was analysed by percentage contribution of segments such as geography. Finally, geographic split of segments was further analysed to arrive at region- and country-wise break-up of the overall market.

Sensors in medical and biotechnology

In medicine and biotechnology, sensors detect specific biological, chemical or physical processes and then transmit or report this data. Some sensors work outside the body, while some are implanted inside the body.

Sensors used in the medical sector range from temperature sensors, pressure detectors, flow sensors, acoustic sensors and gas sensors to cameras, image sensors and magnetic field sensors. Image sensors and cameras used in the medical sector can be optical, X-ray and ultrasonic.

Use of sensors in the medical sector depends strongly on the application. The market for biomedical sensors and cameras as well as image sensors is growing in importance. MEMS sensors are expected to enable the design and production of

Fig. 1: Block diagram of a sensor

